

Name _____ Date _____ Period _____

Parent Functions II: x^3 and $\sqrt[3]{x}$ - Practice

Algebra II | Unit 1 | Lesson 5 Practice | TEK A2.2.A

Objective: I can evaluate x^3 and $\sqrt[3]{x}$, compare their solution counts to squaring/square roots, and name shared attributes.

Section 1 — Recall

1. Evaluate: $(-4)^3 =$ _____ $\sqrt[3]{(-64)} =$ _____
2. Complete: $f(x) = x^3$ at $x = -1, 0, 3$.
 $f(-1) =$ _____ $f(0) =$ _____ $f(3) =$ _____

Section 2 — Apply

3. Solve $x^3 = 64$. How many real answers? Compare to $x^2 = 64$ - how many?
 $x^3 = 64$: _____ $x^2 = 64$: _____
4. Evaluate $\sqrt[3]{343}$ and $\sqrt[3]{(-343)}$.

Section 3 — Challenge

5. A cube-shaped tank has volume $V = x^3$ cubic feet. If $V = 1000 \text{ ft}^3$, find the side length x . Then, purely mathematically, find $\sqrt[3]{(-1000)}$. What does this tell you about cube roots vs. square roots for negative inputs?

 $x =$ _____ $\sqrt[3]{(-1000)} =$ _____
6. Which two of the four parent functions studied so far ($\sqrt{x}$, $|x|$, x^3 , $\sqrt[3]{x}$) accept every real number as both domain and range? Justify using attributes, not equations.

Section 4 — Explain It

7. A classmate says: "Since $x^2 = 25$ has two solutions, $x^3 = 27$ should also have two solutions, because cubing is just like squaring but bigger." Is the classmate correct? Explain using the graphs' symmetry.

